Data Analytics

MIDSEM PRACTICE PAPER | 30 marks | Suggested time: 90 minutes

Modules 1-3

PROVISIONAL mapping: the Teams DA folder is empty. Selections cover descriptive statistics, preparation and statistical testing, based on the source CO labels. The two source papers contain no CO2 questions, so full module-2 coverage cannot be claimed. Use statistical tables where needed.

Answer all questions. Show working and draw labelled diagrams where appropriate. Marks and duration are a practice format, not a confirmed university blueprint.

Question 1 [5 marks]

(i) Find the standard deviation and skewness for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61.
(ii) Give an example of the combining-variables concept in dataset preparation.
(iii) A CMO compares three new cola flavours using ratings from 12 customers on a 1-9 scale. Conduct the appropriate analysis. Ratings: Flavour A: 4, 2, 4, 2; Flavour B: 6, 4, 5, 5; Flavour C: 8, 6, 8, 6.

Source: DA-R | original Q3A

Question 2 [5 marks]

(i) Find the elements of a box plot for: 78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75, 79, 63, 42, 61.
(ii) Why is it sometimes necessary to create subsets from a large dataset before analysis?
(iii) Measure the strength of the relationship between the rankings assigned by two interviewers to candidates A-L. Interviewer 1: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12. Interviewer 2: 1, 2, 4, 3, 6, 5, 8, 7, 10, 9, 12, 11.

Source: DA-M | original Q3A

Question 3 [3 marks]

A medical equipment company claims that its new blood test kit has mean accuracy 95%. A random sample of 36 kits has mean accuracy 94.7%, with known population standard deviation 0.02 (accuracy expressed as a proportion). At the 5% significance level, test whether the claim is justified against a two-sided alternative. Critical z-values: ±1.96.

Source: DA-R | original Q1B

Question 4 [3 marks]

A hospital laboratory claims that a new COVID testing kit correctly identifies infection 95% of the time. A researcher believes the true accuracy is less than 95%. A sample of 50 test results has mean accuracy 0.93; the known population standard deviation is 0.04. Test the claim at the 5% significance level. Critical z-value: -1.645.

Source: DA-M | original Q1B

Question 5 [5 marks]

Test the effectiveness of vaccination in preventing smallpox at the 5% significance level.
Vaccinated: attacked 31, not attacked 469, total 500.
Not vaccinated: attacked 185, not attacked 1315, total 1500.
Overall: attacked 216, not attacked 1784, total 2000.

Source: DA-R | original Q2A

Question 6 [5 marks]

Per-acre wheat production (tons) is recorded for three varieties and four fertilizers, with each combination replicated twice. Test whether variety differences and fertilizer differences are significant at the 5% level. Each pair below gives the two replicates, in fertilizer order F1, F2, F3, F4.
Variety 1: (8,9), (9,8), (6,7), (10,11).
Variety 2: (7,6), (8,9), (5,4), (9,10).
Variety 3: (10,9), (11,12), (8,7), (12,13).

Source: DA-M | original Q2A

Question 7 [2 marks]

A data analyst is studying the annual income levels (in Rs. 000s) of 12 employees in a company: [180, 200, 210, 215, 220, 230, 250, 270, 300, 320, 400, 480] Apply both equal-width and equal-frequency binning with 3 bins. Using histogram analysis, discuss which binning approach (equal-width or equal-frequency) provides a better representation of the income distribution and justify your reasoning.

Source: DA-R | original Q5C

Question 8 [2 marks]

A data scientist is analysing the following cholesterol level (mg/dL) readings of 12 patients: [145, 155, 160, 162, 165, 172, 180, 190, 200, 220, 240, 260]. Apply equal-width and equal-frequency binning with 3 bins. Using histogram analysis, explain which binning approach (width-based or frequency-based) better represents the data distribution and why.

Source: DA-M | original Q5C

Sources and selection notes

Working interpretation: NSM means end-semester. This has not yet been confirmed. Questions come only from the local semester-3 CSE papers, using the exact current subject titles. R = regular; M = makeup. Wording is lightly edited, tables transcribed, original question references retained. No new question scenarios were invented.

DA-R: 2025-26 Regular - CSS_2103-Data Analytics

Local source: /Users/shritaake/Desktop/code/paper/public/authoritative/MIT 2021-26/MIT 2025-26 QPs 3rd March 2026/November and December 2025 (Regular)/MIT Sem 1 to Sem 7 QPs/Semester 3/CSS_2103-Data Analytics.pdf

DA-M: 2025-26 Makeup - CSS_2103-Data Analytics

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